Type MS · Complete a truth table

Answer 1 of 5 · Question 4 part c.i · 3 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part c.i in this mark scheme extract

4(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has high bandwidth/transmission speed (1) • It has low latency (1) • It has low energy consumption (1) • It has high capacity (1) • It has high security (1)	Allow 'more users can be connected' for MP4	**2**
Question	Answer	Additional Guidance	Mark
4(a)(ii)	Award up to **two** marks for a linked description such as: • Biometric identifiers (fingerprint, iris, face) (1) matched against those stored in the app/database (1) • Two-factor authentication / 2FA (1) requiring users to enter a code delivered to a separate system (1) • Users can be asked for memorable information/security question (1) that only they will know the answer to (1)	System in MP2b may be on a separate device or a separate app	**2**
Question	Answer	Additional Guidance	Mark
4(a)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • To identify any bad programming practices/shortcuts/bugs (1) • To highlight code vulnerabilities / enhance security (1) • To make sure code is robust/will deal with exceptions (1) • To make sure that good programming practices/company policies are maintained (1) • To assess the efficiency of the code (1) • To improve code quality (1) • To share good practice/new techniques (1)		**2**

Type MS · Complete a truth table

Answer 1 of 5 · Question 4 part c.i · 3 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part c.i in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(b)</td><td>Award one mark for each correct row:
<table>
<tr><th></th><th>8-bit binary pattern</th><th>Denary value</th></tr>
<tr><td>Lowest value</td><td>1111 1111</td><td>-127</td></tr>
<tr><td>Highest value</td><td>0111 1111</td><td>(+)127</td></tr>
</table></td><td></td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(i)</td><td>Award one mark for each correct column:
<table>
<tr><th>O</th><th>S</th><th>P</th><th>O AND S</th><th>NOT P</th><th>(O AND S) AND (NOT P)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table></td><td></td><td>3</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(ii)</td><td>Award up to two marks for a linked explanation such as:
ROM is non-volatile/permanent (1) therefore it retains its contents when the power is switched off / data won't be lost when power is lost (1)
The contents of ROM cannot be changed (1) which makes it suitable for storing the manufacturer's instructions (1)</td><td>Explanation can be a combination of different marking points. E.g. MP1a and MP2b
Do not accept MP1a and MP2a or MP1b and MP2b</td><td>2</td></tr>
<tr><td colspan="3">Total for question 4</td><td>13</td></tr>
</table>

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Complete a truth table

Answer 1 of 5 · Question 4 part c.i · 3 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part c.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Complete a truth table

Answer 2 of 5 · Question 5 part a · 3 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

5(a)	<table><tr><th>P</th><th>Q</th><th>R</th><th>Q OR R</th><th>P AND (Q OR R)</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> Award **one** mark for each of: • All possible combinations in columns P, Q and R (1) • Correct values in Q or R (1) • Correct values in final column (1)	Allow follow through for incorrect mark point 2.	3

Question Number	Answer	Additional Guidance	Mark
5(b)	B AND S AND NOT L Award **one** mark for each of: NOT L or B AND S (1) Fully correct (1)	• Any order and ignore brackets	2

Question Number	Answer	Additional Guidance	Mark
5(c)	Any **two** of: • Remove code vulnerabilities in programming languages (1). • Eliminate bad programming practices (1).		2

Question Number	Answer	Additional Guidance	Mark
5(d)	Award **one** mark each for any of: • Copyright (1) • Patent (1) • Licensing (1) • Trademark (1)		3

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Complete a truth table

Answer 2 of 5 · Question 5 part a · 3 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

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Type MS · Complete a truth table

Answer 2 of 5 · Question 5 part a · 3 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Complete a truth table

Answer 3 of 5 · Question 4 part c · 3 marks

2021 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

Number		Guidance	
4(a)	Award **one** mark from: • Predict/model outcomes (1) • Investigate hypotheses (1) • Run experiments that cannot be run / are dangerous in real life (1) • Simulations can be speeded up/slowed down (1)		1

Type MS · Complete a truth table

Answer 3 of 5 · Question 4 part c · 3 marks

2021 JUN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **two** marks from: • Simulations (may) use/collect large amounts of data (1) • Simulations (may) involve a large number of calculations (1) • Simulations may need more CPU cycles as the amount of data grows (1) • To be of value, the results must be calculated as quickly as possible (1) • Calculations may grow exponentially / become more complex as the simulation runs/is modified (1) • Simulation may output complex graphics (e.g. virtual world) (1)		2

Question Number	Answer	Additional Guidance	Mark
4(c)	<table><tr><th>NOT S</th><th>R AND W</th><th>(NOT S) OR (R AND W)</th></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr></table> Award **one** mark for each of: • Correct values in NOT S column (1) • Correct values in R AND W column (1) • Correct values in final column (1)	Allow follow through for incorrect mark point 1 and/or mark point 2	3

Type MS · Complete a truth table

Answer 3 of 5 · Question 4 part c · 3 marks

2021 JUN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(d)(i)</td><td>Award two marks for a linked description such as:

• Instructions (and data) are stored in (main) memory (1)
• Instructions (and data) are fetched, decoded, and executed (in a sequence by the CPU) (1)</td><td>Must include storage and execution for 2 marks</td><td>2</td></tr>
<tr><td>4(d)(ii)</td><td>Award two marks from:

• Program counter / PC (1)
• Current instruction (register) / CIR (1)
• Memory address (register) / MAR (1)
• Memory buffer/data (register) /MBR/MDR (1)
• Accumulator/ACC (1)</td><td></td><td>2</td></tr>
<tr><td>4(d)(iii)</td><td>Award two marks for a linked description such as:

Address of instruction carried on address bus (to main memory) (1) contents at that address carried back (to the CPU) on the data bus (1)</td><td></td><td>2</td></tr>
<tr><td>4(d)(iv)</td><td>Award two marks from:

<table><tr><th>Change made</th><th>Effect</th></tr><tr><td>Increasing the width of the data bus</td><td>Increases the number of bits/data/ size of word that can be transferred at one time / increases rate of data transfer (1)</td></tr><tr><td>Increasing the width of the address bus</td><td>Increases the number of memory addresses / increases the maximum amount of memory that can be addressed (1)</td></tr></table></td><td>Do not accept increases bandwidth without further explanation</td><td>2</td></tr>
</table>

Type MS · Complete a truth table

Answer 3 of 5 · Question 4 part c · 3 marks

2021 JUN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Complete a truth table

Answer 4 of 5 · Question 5 part d.i · 3 marks

2023 JUN · Paper 01 · extract 1 of 8 · find part d.i in this mark scheme extract

<table>
<tr>
<td>5(a)</td>
<td>Award one mark for each correct cell:

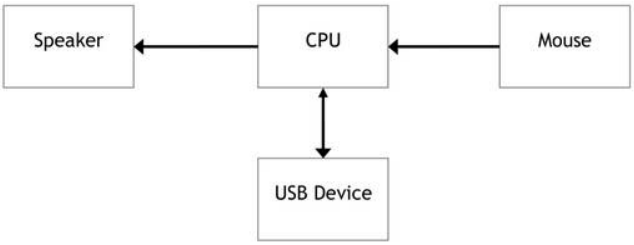
<table>
<tr><th>Req.</th><th>Example</th></tr>
<tr><td>Input</td><td>1. Enter the direction the bee will face (1)
2. Enter the number of steps that the bee will move (1)</td></tr>
<tr><td>Process</td><td>Calculate the path the bee will move along to its new position</td></tr>
<tr><td>Output</td><td>3. The bee appears in its new position (1)
4. The bee flies to its new position (1)
5. The bee faces its new direction (1)</td></tr>
</table>
</td>
<td>Allow direct instructions as input e.g. ‘Turn East’ and consequences for output e.g. ‘Bee turns to face East’ if clear.

The input/output do not have to be directly linked. E.g. Input Turn West, Output Has Moved 20 steps</td>
<td>2</td>
</tr>
</table>

Type MS · Complete a truth table

Answer 4 of 5 · Question 5 part d.i · 3 marks

2023 JUN · Paper 01 · extract 2 of 8 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5b(i)	Award one mark for each: • Line between Mouse and CPU with arrow at CPU end only (1) • Line between CPU and Speaker with arrow at Speaker end only (1) • Line between CPU and USB Device with arrows on both ends (1) 	Allow two independent back/forth arrows between CPU and USB in place of a single bi-directional arrow. Ignore any additional lines.	3

Type MS · Complete a truth table

Answer 4 of 5 · Question 5 part d.i · 3 marks

2023 JUN · Paper 01 · extract 3 of 8 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(ii)	**The only correct answer is B** ***A*** *is not correct because a* ***firewall*** *is an application that is loaded after the device is booted* ***C*** *is not correct because* ***the operating system*** *is loaded after the device is booted* ***D*** *is not correct because* ***the user interface*** *is loaded after the device is booted*		1

Question Number	Answer	Additional Guidance	Mark
5(b)(iii)	Award up to **two** marks for a linked explanation, such as: • Cache makes up for the difference in speed of two devices (1) because it is very high-speed memory (1) • Cache is high-speed memory (1) which acts as a buffer between a faster and a slower device (1) • It speeds up processing (1) because active/commonly used instructions and data are stored nearer to the CPU / the CPU doesn't have to wait while instructions/data are fetched from RAM (1)		2

Type MS · Complete a truth table

Answer 4 of 5 · Question 5 part d.i · 3 marks

2023 JUN · Paper 01 · extract 4 of 8 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)	Award up to **two** marks for a linked description, such as: • Uses a sensor (1) to check the water level in the machine (1) • Uses a thermometer (1) to check the water temperature (1) • Uses a timer (1) to control the time required for the wash cycle/rinse/drain/spin (1) • Uses a motor (1) to rotate the drum (1)	The first mark is for identification and the second is for the linked description. It is possible to gain one mark for a description without identifying the sensor/actuator.	2

Type MS · Complete a truth table

Answer 4 of 5 · Question 5 part d.i · 3 marks

2023 JUN · Paper 01 · extract 5 of 8 · find part d.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award one mark for each two correct rows.

<table>
<tr><th>A</th><th>B</th><th>C</th><th>A AND C</th><th>(A AND C) OR B</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td></td><td>3</td></tr>
</table>

Type MS · Complete a truth table

Answer 4 of 5 · Question 5 part d.i · 3 marks

2023 JUN · Paper 01 · extract 6 of 8 · find part d.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(ii)</td><td>Award one mark for each of:

• M OR P (1)
• NOT C (1)
• <expr1> AND <expr2> (1)

Example statements:

(M OR P) AND (NOT C)
(M OR P) AND NOT C
(M AND NOT C) OR (P AND NOT C)
M AND NOT C OR P AND NOT C

For information only:

<table>
<tr><th>M</th><th>P</th><th>C</th><th>M OR P</th><th>NOT C</th><th>(M OR P) AND (NOT C)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td><td>Third bullet awarded even if first two are not

Order of precedence is NOT, AND, OR

Check brackets carefully, if provided</td><td>3</td></tr>
</table>

Type MS · Complete a truth table

Answer 4 of 5 · Question 5 part d.i · 3 marks

2023 JUN · Paper 01 · extract 7 of 8 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. The operating system uses a scheduler to control processes (1) 2. The operating system holds processes in a queue (1) 3. Some processes may be given higher priorities than others (1) 4. Each process gains accesses to the CPU for a short time / time slice to execute (1) 5. Processes are swapped to/from (the queue / CPU)	Do not award responses about memory management, as the question is about sharing the CPU only	2

Type MS · Complete a truth table

Answer 4 of 5 · Question 5 part d.i · 3 marks

2023 JUN · Paper 01 · extract 8 of 8 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 1 of 13 · find part a in this mark scheme extract

<table>
<tr>
<td>3(a)</td>
<td>A1
A2
A3
A4</td>
<td>Award one mark for correct column (A OR B)
Award one mark for correct column (NOT B)
Award one mark for first set of two cells in third column
Award one mark for second set of two cells in third column

<table>
<tr><th>A</th><th>B</th><th>A OR B</th><th>NOT B</th><th>(A OR B) AND (NOT B)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td>
<td>Award follow through on last column only

5th column can be judged
• On columns 1 and 2 for MP3 and MP4
or
• as FT from columns 3 and 4 for MP3 and MP4
• award highest mark</td>
<td>4</td>
</tr>
</table>

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 2 of 13 · find part a in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 3 of 13 · find part a in this mark scheme extract

3(b)		Award **one** mark for each of:		
	B1	Correct output taken from array (1)	Must be from array i.e. Indexing used	
	B2	Use of iteration to loop over every item in the array (1)	Award foreach loops Accept any array	
	B3	Use of appropriate method to step accurately, i.e. every other item starting at position 0 (1)	Any appropriate step method Not hard coded index	**3**

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 4 of 13 · find part a in this mark scheme extract

C#

```
// Q03bFINISHED

String[] starNames = {"Alasia", "Beid", "Castor", "Denebola",
                      "Electra", "Fafnir", "Gudja", "Haedus",
                        "Izar", "Jishui", "Kang", "Lich",
                        "Maia", "Nahn", "Ogma", "Peacock"};


// Write your code here
int start = 0;
int stop = starNames.Length;
int step = 2;

for (int index = start; index < stop; index = index + step)
{
    Console.WriteLine (starNames[index]);
}

// Alternative
for (int i=0; i<starNames.Length; i=i+2)
{
    Console.WriteLine (starNames[i]);
}

```

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 5 of 13 · find part a in this mark scheme extract

Java

```java
// Q03bFINISHED


public class Q03bFINISHED
{
    public static void main (String[] args)
    {
        String starNames[] = {"Alasia", "Beid", "Castor", "Denebola",
                              "Electra", "Fafnir", "Gudja", "Haedus",
                              "Izar", "Jishui", "Kang", "Lich",
                              "Maia", "Nahn", "Ogma", "Peacock"};

        // Write your code here
        int start = 0;
        int stop = starNames.length;
        int step = 2;

        for (int index = start; index < stop; index = index + step)
        {
            System.out.println (starNames[index]);
        }

        // Alternative
        for (int i=0; i<starNames.length; i=i+2)
        {
            System.out.println (starNames[i]);
        }

    }
}
```

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 6 of 13 · find part a in this mark scheme extract

Python

```python
# Q03bFINISHED

starNames = ["Alasia", "Beid", "Castor", "Denebola",
             "Electra", "Fafnir", "Gudja", "Haedus",
             "Izar", "Jishui", "Kang", "Lich",
             "Maia", "Nahn", "Ogma", "Peacock"]

# Write your code here

start = 0
stop = len (starNames)
step = 2

for index in range (start, stop, step):
    print (starNames[index])

# Alternative
for index in range (0, len (starNames), 2):
    print (starNames[index])
```

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 7 of 13 · find part a in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 8 of 13 · find part a in this mark scheme extract

3(c)		Award **one** mark for each:		
	C1	Take a string input (1)	• `aString = Console.ReadLine ();` • `aString = myKeyboard.nextLine();` • `aString = input ("Enter a number: ")`	**6**
	C2	Complete the selection with 'if' (1)		
	C3	Convert string to integer (1)	• `aNum = Convert.ToInt32 (aString);` • `aNum = Integer.parseInt (aString);` • `aNum = int (aString)`	
	C4	Any one correct set of relational operators (1) ignore logical operator first two tests ignore test dealing with >= 1	Python • `(aNum <= 20) or (aNum >= 60)` • `(aNum < 21) or (aNum > 59)` • `(aNum > 30) and (aNum < 40)` • `(aNum >= 31) and (aNum <= 39)` C# and Java • `((aNum <= 20) || (aNum >= 60))` • `(aNum < 21) || (aNum > 59))` • `((aNum > 30) && (aNum < 40))` • `((aNum >= 31) && (aNum <= 39))`	
	C5	Both logical operators used correctly in each test (1) Ignore relational operators First test: **or** (must attempt to address 20 and 60 with relational, even if C4 not awarded Second test: **and**		
	C6	Complete the third test (1)	All Languages: `(aNum == 30)`	

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 9 of 13 · find part a in this mark scheme extract

C#

```
// Q03cFINISHED

String aString = "";
int aNum = 0;

Console.WriteLine ("Enter a number:");

// Complete the line to take a string input
aString = Console.ReadLine ();

// Complete the selection statement
if (aString != "")
{
    // Convert the string to an integer
    aNum = Convert.ToInt32 (aString);
    if (aNum > 0)
    {
        // Complete the test
        if ((aNum <= 20) || (aNum >= 60))
        {
            Console.WriteLine ("Acceptable");
        }
        // Complete the test
        else if ((aNum >30) && (aNum < 40))
        {
            Console.WriteLine ("Centre");
        }
        // Complete the test
        else if (aNum == 30)
        {
            Console.WriteLine ("Perfect");
        }
        else
        {
            Console.WriteLine ("No message");
        }
    }
    else
    {
        Console.WriteLine ("The number must be greater than zero");
    }
}
else
{
```

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 10 of 13 · find part a in this mark scheme extract

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 11 of 13 · find part a in this mark scheme extract

Java

```java
// Q03cFINISHED

import java.util.Scanner;

public class Q03cFINISHED
{
    public static void main (String[] args)
    {
        String aString = "";
        int aNum = 0;
        Scanner myKeyboard = new Scanner (System.in);

        System.out.println ("Enter a number:");

        // Complete the line to take a string input
        aString = myKeyboard.nextLine();

        // Complete the selection statement
        if (aString != "")
        {
            // Convert the string to an integer
            aNum =  Integer.parseInt (aString);
            if (aNum > 0)
            {
                // Complete the test
                if ((aNum <= 20) || (aNum >= 60))
                {
                    System.out.println ("Acceptable");
                }
                // Complete the test
                else if ((aNum >30) && (aNum < 40))
                {
                    System.out.println ("Centre");
                }
                // Complete the test
                else if (aNum == 30)
                {
                    System.out.println ("Perfect");
                }
                else
                {
                    System.out.println ("No message");
                }
            }
            else
            {
                System.out.println ("The number must be greater than zero");
            }
        }
```

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 12 of 13 · find part a in this mark scheme extract

Python

```python
# Q03cFINISHED

# Complete the line to take a string input
aString = input ("Enter a number: ")

# Complete the selection statement
if (aString != ""):

    # Convert the string to an integer
    aNum = int (aString)

    if (aNum > 0):

        # Complete the test
        if (aNum <= 20) or (aNum >= 60):
            print ("Acceptable")
        # Complete the test
        elif (aNum > 30) and (aNum < 40):
            print ("Centre")
        # Complete the test
        elif (aNum == 30):
            print ("Perfect")
        else:
            print ("No message")
    else:
        print ("The number must be greater than zero")
else:
    print ("You must provide a number")
```

Type MS · Complete a truth table

Answer 5 of 5 · Question 3 part a · 4 marks

2024 JUN · Paper 02 · extract 13 of 13 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark